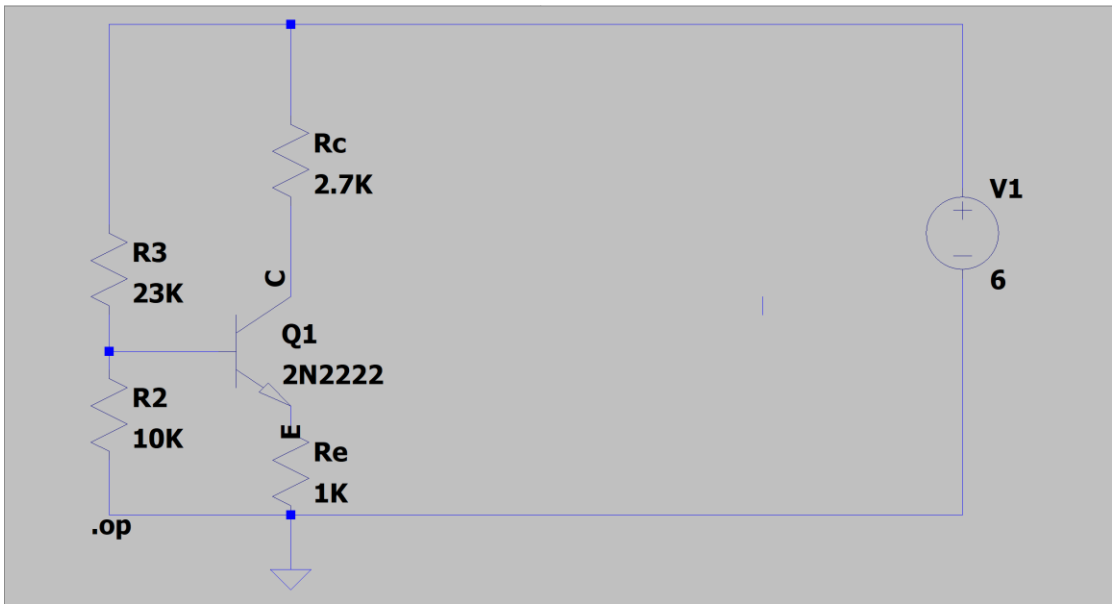
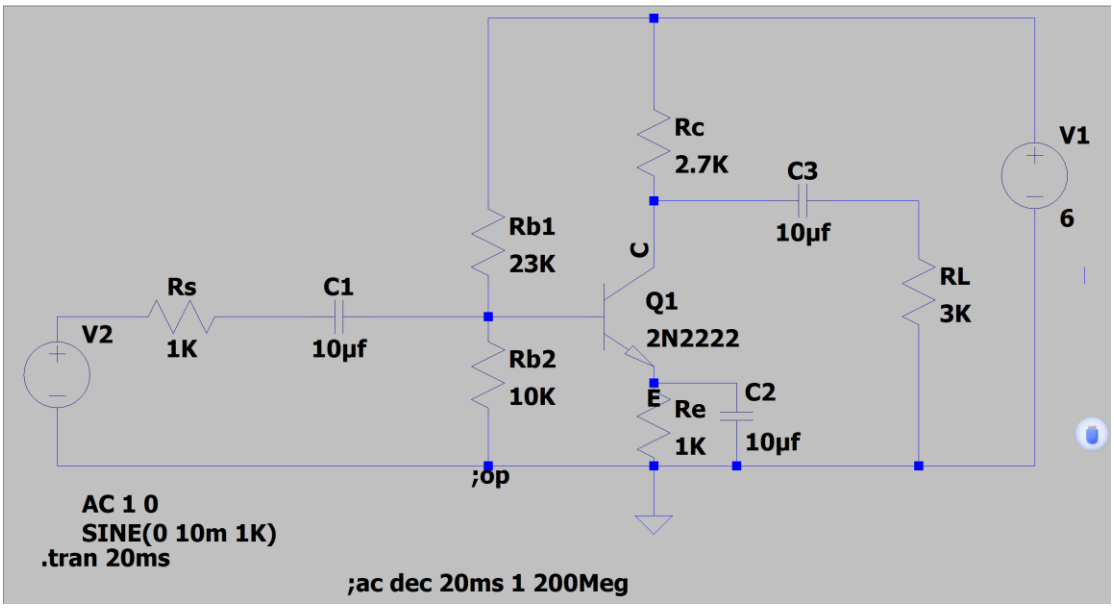
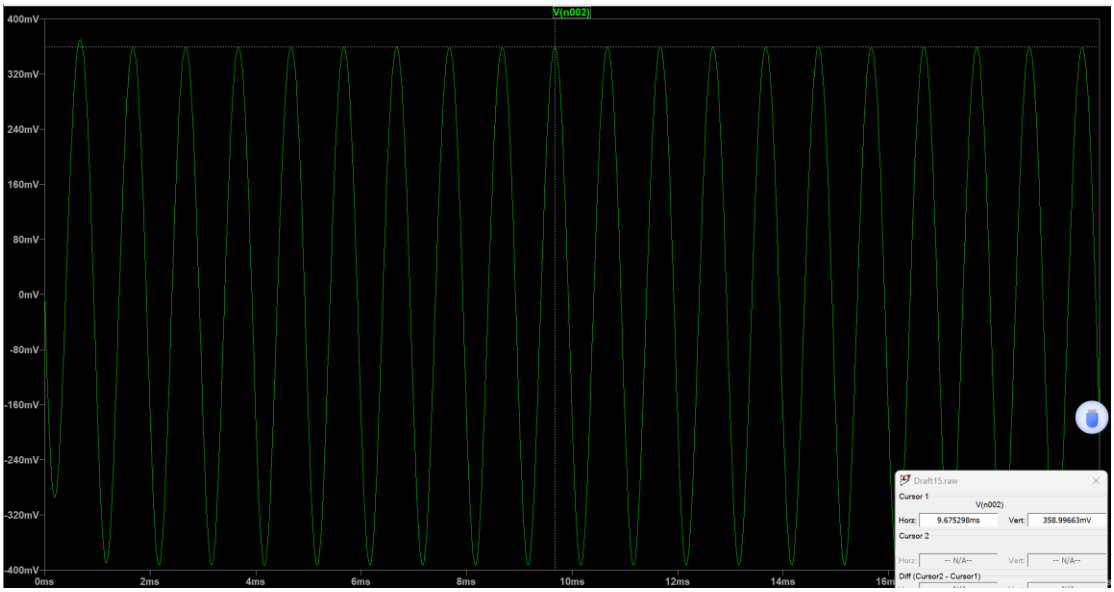



```
       --- Operating Point ---

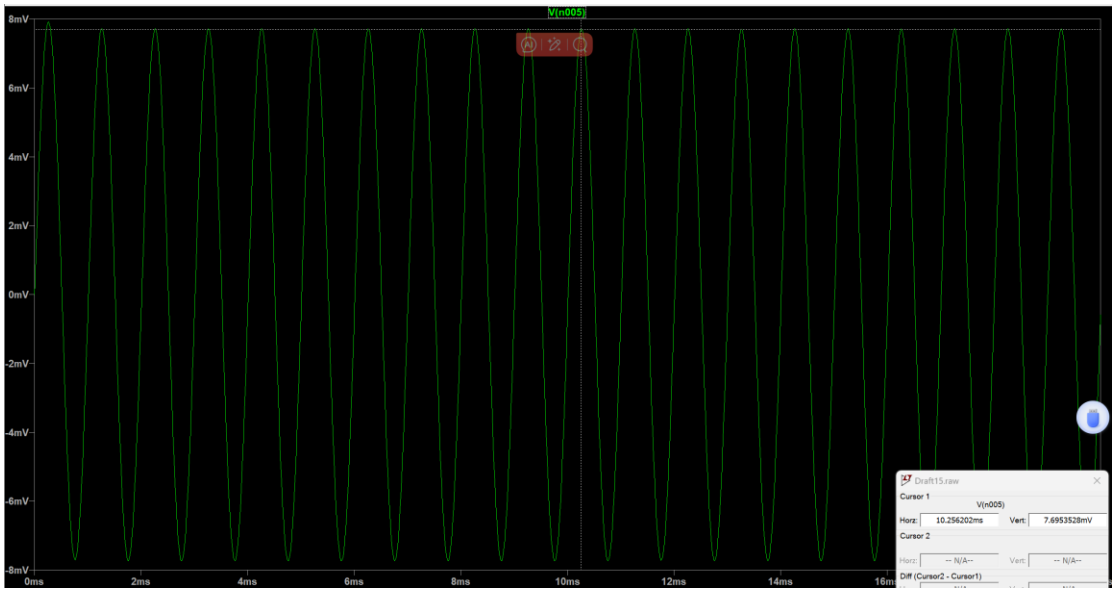
V(n003):        1.77961       voltage
V(n001):        6             voltage
V(c):           2.98665       voltage
V(e):           1.12159       voltage
Ic(Q1):         0.00111605    device_current
Ib(Q1):         5.534e-006    device_current
Ie(Q1):         -0.00112159   device_current
I(Re):          0.00112159    device_current
I(Rc):          0.00111605    device_current
I(R3):          0.000183495   device_current
I(R2):          0.000177961   device_current
I(V1):          -0.00129955   device_current
```

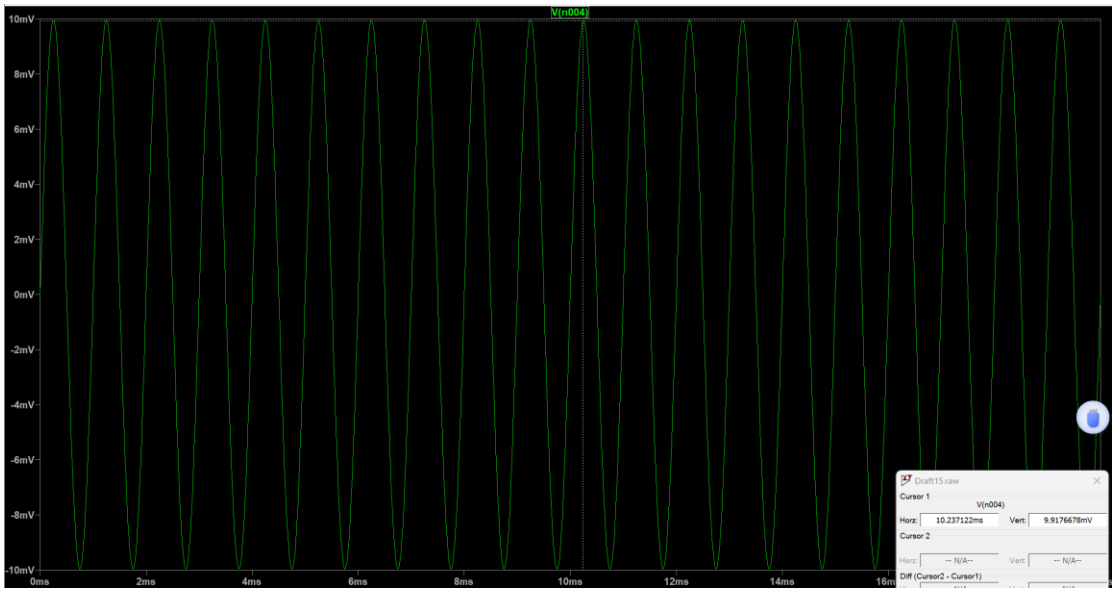
$V_{CEQ}$=Vc-Ve=(2.98665-1.12159)V=1.86506V



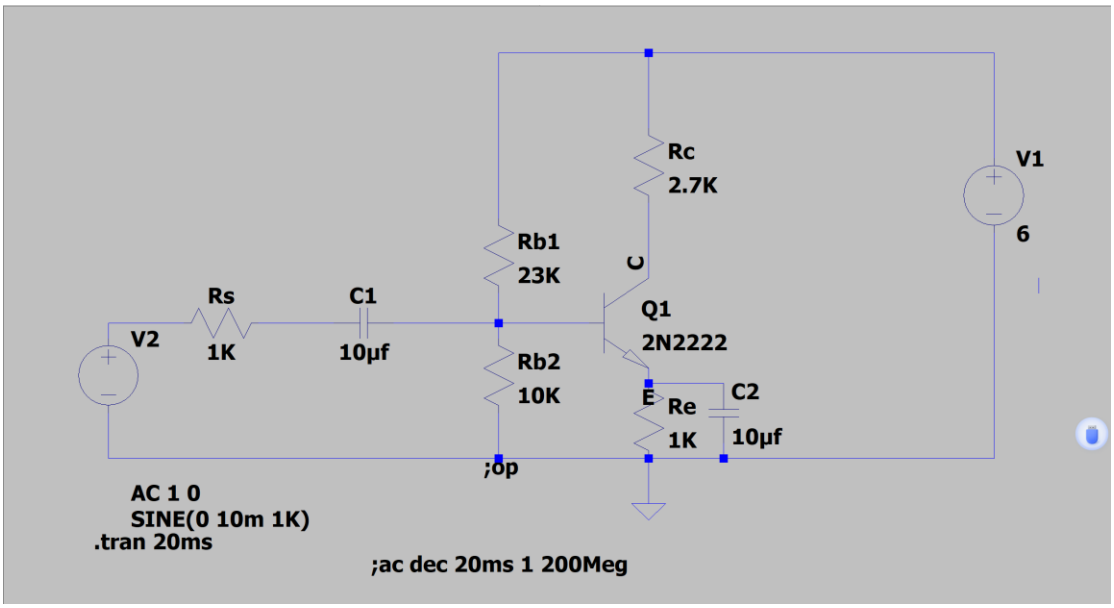

$$A_v = \frac{V_o}{V_i} = \frac{358mV}{10mV} \approx 35.8$$



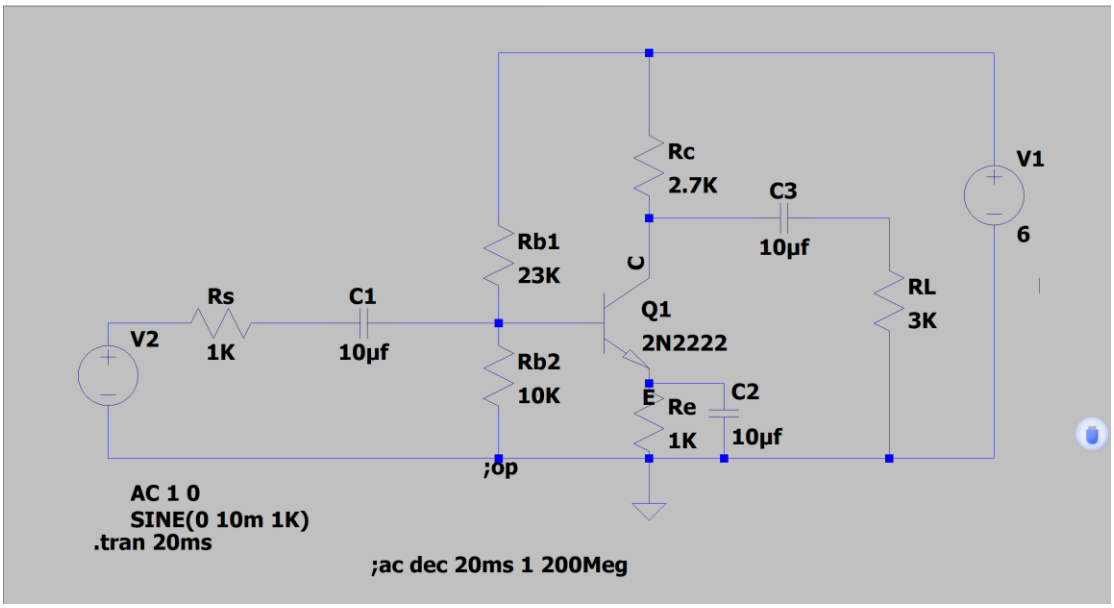

$$R_i = \frac{V_i}{I_i} = \frac{V_i}{V_s - V_i} R_s = \frac{7.89mV}{9.91mV - 7.89mV} 1\text{K}\Omega = 3.9\text{K}$$

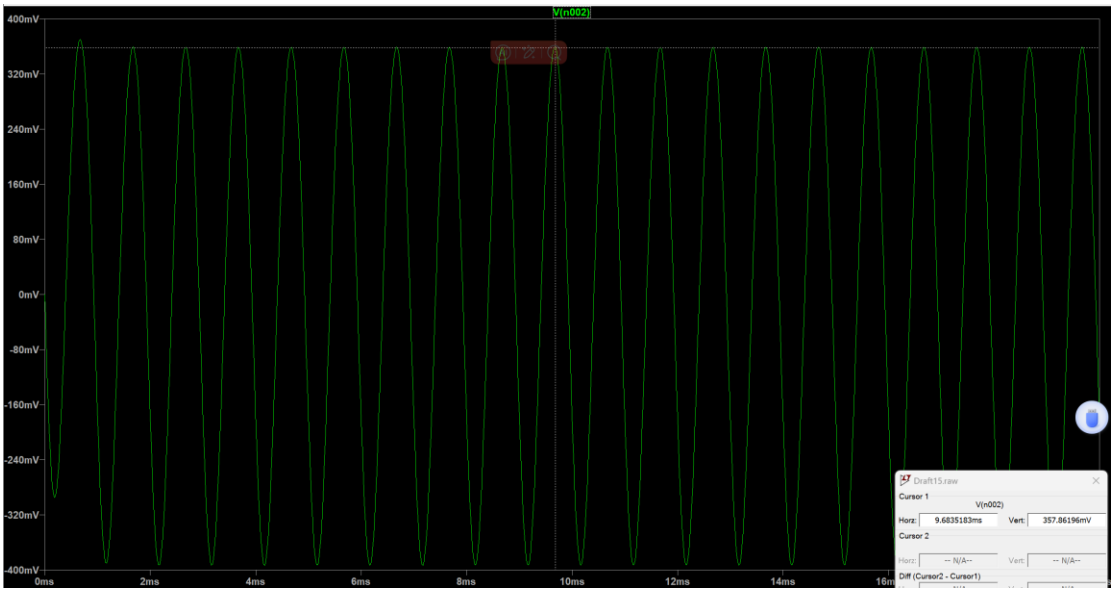
V0:


Rc
2.7K
V1
6
Rb1
23K
C
Rs
C1
Q1
2N2222
V2
1K
10µf
Rb2
10K
E
Re
C2
1K
10µf
;op
AC 1 0
SINE(0 10m 1K)
.tran 20ms
;ac dec 20ms 1 200Meg

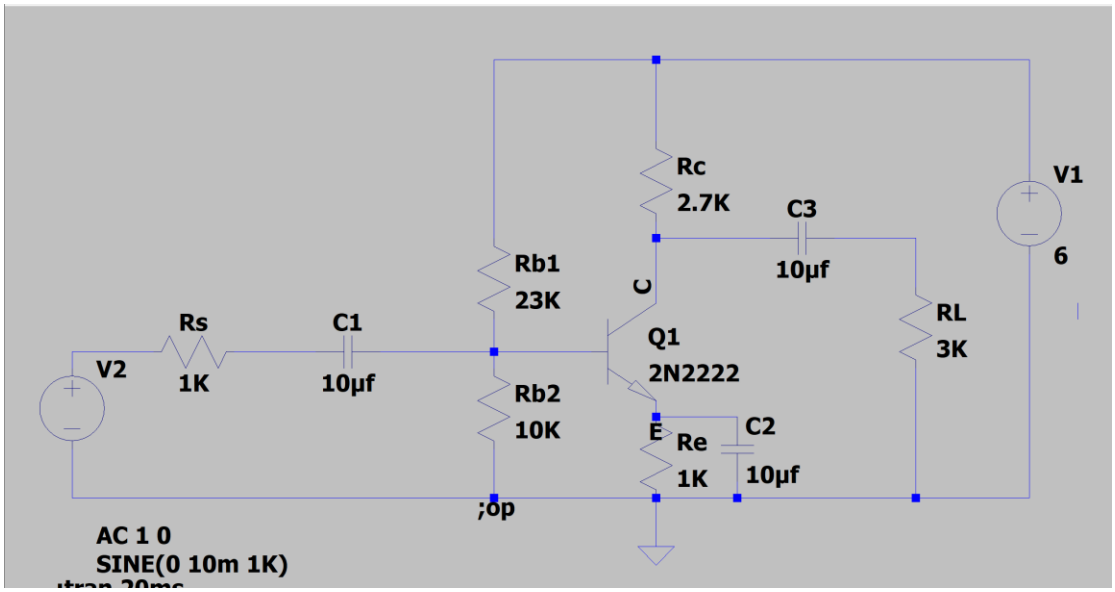

V(nc_01)
700mV
600mV
500mV
400mV
300mV
200mV
100mV
0mV
-100mV
-200mV
-300mV
-400mV
-500mV
-600mV
-700mV
-800mV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
16ms
Draft15.raw
Cursor 1
V(nc_01)
Horz: 9.6917386ms
Vert: 671.85545mV
Cursor 2
Horz: -- N/A--
Vert: -- N/A--
Diff (Cursor2 - Cursor1)

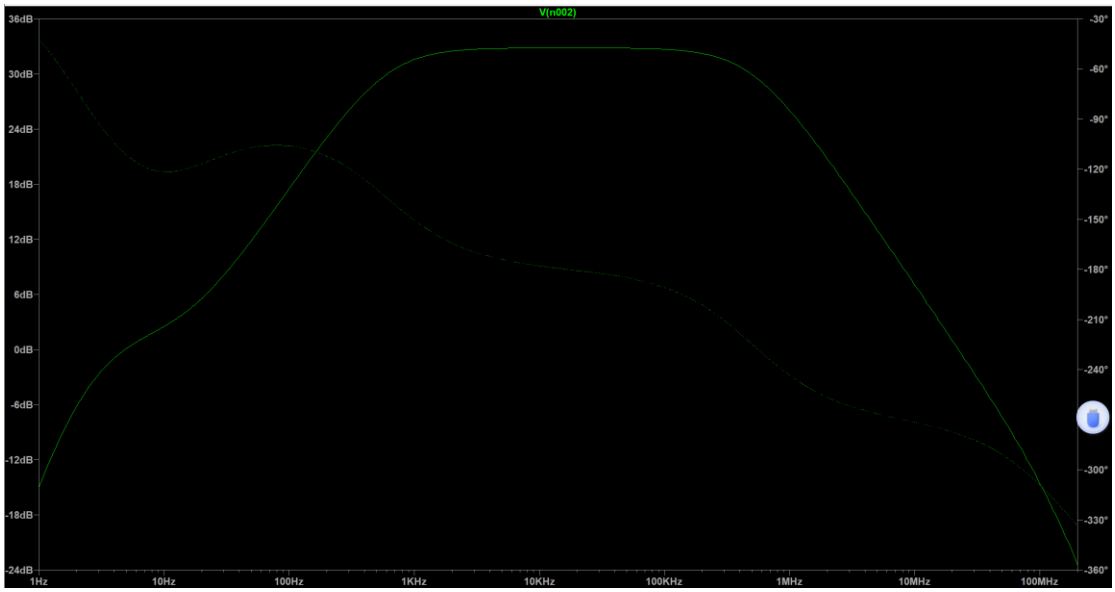
VL:




$$R_o = \left(\frac{V_o}{V_L} - 1\right) R_L \qquad = \left(\frac{671mV}{357mV} - 1\right) 3K\Omega \;\; = 2.64\text{k}\Omega$$




$f_h$=583HZ

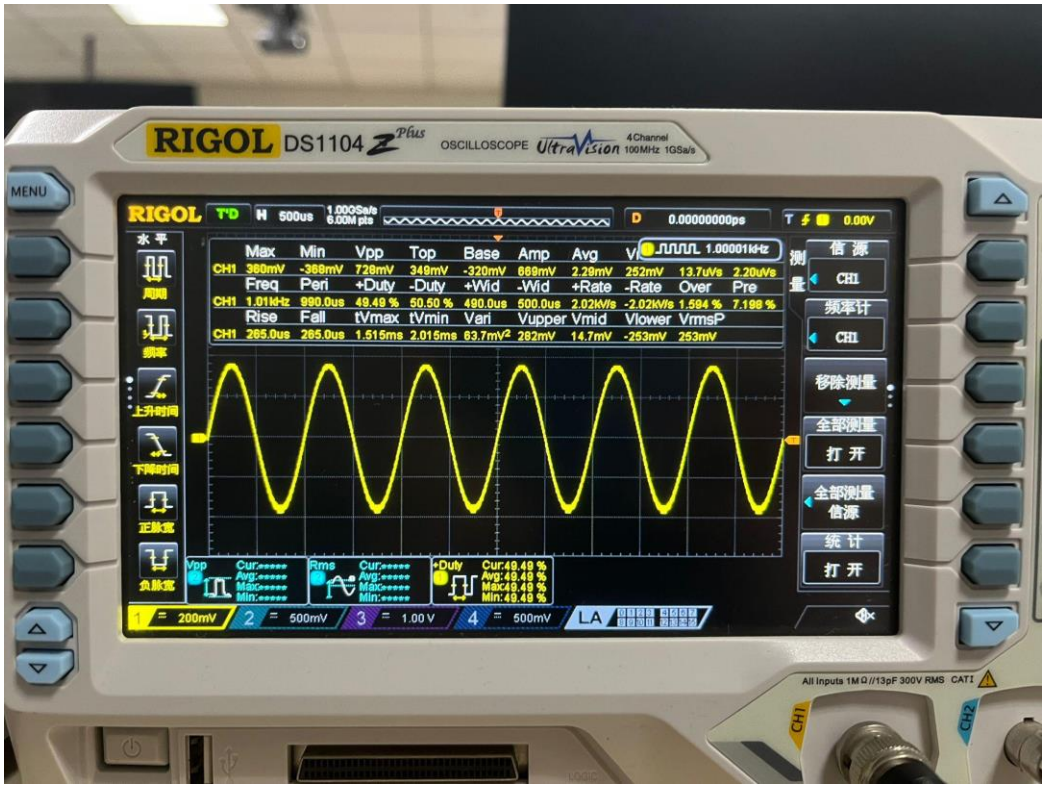
$f_L$=450KHZ

Fill the result in below table

| quiescent point | voltage gain | input resistance | output resistance | Passband |
|---|---|---|---|---|
| 1.86506V | $A_v = \frac{V_o}{V_i}$ $= -\frac{358mV}{10mV} \approx$ -35.8 | 3.9KΩ | 2.64KΩ | $f_h$=583HZ $f_L$=450KHZ |

**实物：**

16V
Tantalum


RIGOL DS1104 Z Plus
OSCILLOSCOPE UltraVision 4Channel 100MHz 1GSa/s
MENU
RIGOL
H 500us
D 0.00000000ps
0.00V
Max Min Vpp Top Base Amp Avg
CH1 360mV -368mV 728mV 349mV -320mV 669mV 2.29mV 252mV 13.7uVs 2.20uVs
Freq Peri +Duty -Duty +Wid -Wid +Rate -Rate Over Pre
CH1 1.011kHz 990.0us 49.49 % 50.50 % 490.0us 500.0us 2.02kV/s -2.02kV/s 1.594 % 7.198 %
Rise Fall tVmax tVmin Vari Vupper Vmid Vlower VrmsP
CH1 265.0us 265.0us 1.515ms 2.015ms 83.7mV² 282mV 14.7mV -253mV 253mV
1.00001kHz
测量
信源
CH1
频率计
CH1
移除测量
全部测量
打开
全部测量信源
统计
打开
水平
周期
频率
上升时间
下降时间
正脉宽
负脉宽
Vpp Cur:***** Avg:***** Max:***** Min:*****
Rms Cur:***** Avg:***** Max:***** Min:*****
Duty Cur:49.49 % Avg:49.49 % Max:49.49 % Min:49.49 %
1 200mV
2 500mV
3 1.00 V
4 500mV
LA
All Inputs 1MΩ//13pF 300V RMS CAT I
CH1
CH2